



Sydney Girls High School

2013

TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION

Extension 1

Mathematics

General Instructions

- Reading Time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11 – 14, show relevant mathematical reasoning and/or calculations

Total marks – 70

Section I Pages 3 – 6

10 Marks

- Attempt Questions 1 – 10
- Answer on the Multiple Choice answer sheet provided
- Allow about 15 minutes for this section

Section II Pages 7 – 13

60 Marks

- Attempt Questions 11 – 14
- Answer on the blank paper provided
- Begin a new page for each question
- Allow about 1 hours and 45 minutes for this section

Name:

Teacher:

THIS IS A TRIAL PAPER ONLY

It does not necessarily reflect the format or the content of the 2013 HSC Examination Paper in this subject.

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Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10

(1) The acute angle between the straight lines $y = \sqrt{3}x + 2$ and $y = 2$ is :

(A) 30°

(B) 60°

(C) 47°

(D) 68°

(2) The value of $\lim_{n \rightarrow \infty} \frac{5(10^n) + 3}{2(10^n) + 5}$ is:

(A) $\frac{3}{5}$

(B) 0

(C) 1

(D) $\frac{5}{2}$

(3) The exact value of k given $\int_0^1 \frac{dx}{x^2 + 3} = k\pi$ is:

(A) $\sqrt{3}$

(B) $\frac{\sqrt{3}}{9}$

(C) $\frac{\sqrt{3}}{18}$

(D) $6\sqrt{3}$

(4) Which of the following is the derivative of $x^2 \cos^{-1} 3x$?

(A) $2x \sin^{-1} 3x$

(B) $2x \cos^{-1} 3x + x^2 \sin^{-1} 3x$

(C) $2x \cos^{-1} 3x - \frac{x^2}{\sqrt{1-9x^2}}$

(D) $2x \cos^{-1} 3x - \frac{3x^2}{\sqrt{1-9x^2}}$

(5) The solution to $\ln(x^3 + 19) = 3 \ln(x + 1)$ is:

(A) $x = -3$ or $x = 2$

(B) $x = 3$

(C) $x = -2$

(D) $x = 2$

(6) The exact value of $\int_0^{\frac{\pi}{4}} \cos^2 \frac{1}{2}x \, dx$ is :

(A) $\frac{1+\pi}{\sqrt{2}}$

(B) $\frac{2\sqrt{2}+\pi}{8}$

(C) $\frac{2\sqrt{2}+\pi}{4}$

(D) $\frac{\sqrt{2}+\pi}{8}$

(7) The domain of $y = \cos^{-1} \sqrt{\frac{1}{4} - x^2}$ is :

(A) $0 \leq x \leq \frac{1}{2}$

(B) $\frac{-1}{4} \leq x \leq \frac{1}{2}$

(C) $\frac{-1}{2} \leq x \leq \frac{1}{2}$

(D) $\frac{1}{4} \leq x \leq \frac{1}{2}$

(8) A metal disc , 5 cm radius , expands when heated. If the radius is increasing at the rate of 0.01 cm / sec , the rate at which the area of one of the faces is increasing is given by:

(A) $\frac{\pi}{10} \text{ cm}^2 / \text{sec}$

(B) $\frac{\pi}{5} \text{ cm}^2 / \text{sec}$

(C) $\frac{2\pi}{5} \text{ cm}^2 / \text{sec}$

(D) $\frac{5\pi}{2} \text{ cm}^2 / \text{sec}$

(9) Two roots of the equation $x^3 - 2x^2 + kx + 18 = 0$ are opposites. The value of k is :

(A) -9

(B) 9

(C) -6

(D) 6

(10) A point moving with simple harmonic motion starts from a point 5cm from the centre of the motion with a speed of 1cm / s . The period is 8 seconds. The maximum acceleration is:

(A) 4.9ms^{-2}

(B) 5.2ms^{-2}

(C) 24.4ms^{-2}

(D) 25.6ms^{-2}

Section II

60 marks

Attempt Questions 11 – 14

Allow about 1 hours and 45 minutes for this section

Answer on the blank paper provided. Begin a new page for each question

Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (Begin a New Page)

(15 Marks)

(a) By making the substitution $u^2 = x + 1$, find $\int \frac{x+2}{\sqrt{x+1}} dx$ [2]

(b) Solve : $x + 2 < \frac{4}{x-1}$ ($x \neq 1$) [3]

(c) Find the general solution (in radian form) of the equation $\cos 2x = \cos x$ [3]

(d) i) Sketch the graph of the curve $y = 3 \sin^{-1}(x/2)$, clearly indicating the domain and range. [1]

ii) Find the area enclosed between the curve $y = 3 \sin^{-1}(x/2)$, the line $x = 1$ and the positive x axis. [3]

(e) Consider the series $\tan x + \tan^3 x + \tan^5 x + \dots$, where $0 \leq x \leq \frac{\pi}{4}$

i) Explain why this series has a limiting sum [1]

ii) Show that $S_{\infty} = \frac{1}{2} \tan 2x$ [2]

End of Question 11

Question 12 (Begin a New Page)

(15 Marks)

- (a) Use mathematical induction to show that $5^n + 2(11^n)$ is a multiple of 3 for all positive integers n . [3]

- (b) At time t minutes the number of individuals in each of population

A and B is given by $N_A = 15 + 20e^{-0.5t}$ and $N_B = 5 + 40e^{-0.5t}$ respectively.

- i) Find the initial size of population A [1]
- ii) Find the initial rate of change of population B [1]
- iii) Find the time at which the two population sizes are equal. [2]

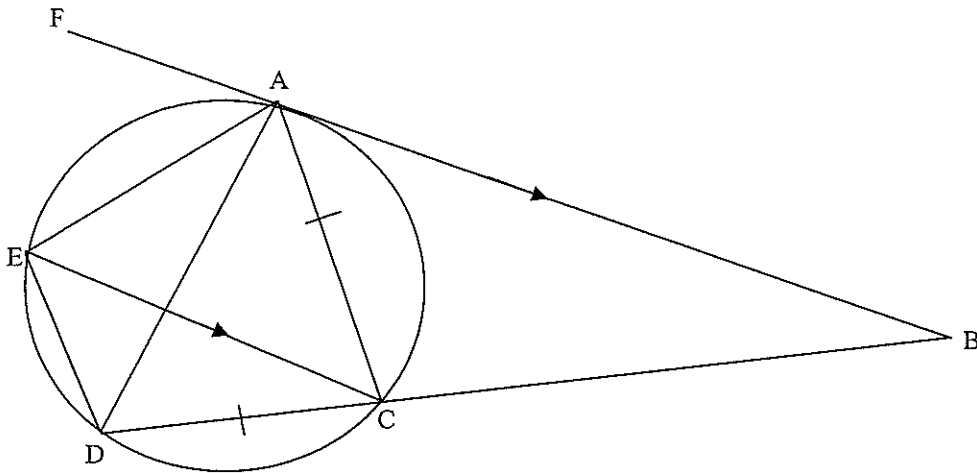
- (c) A particle moves along the x axis according to the equation

$$x = 6 \sin 2t - 2\sqrt{3} \cos 2t .$$

- i) Express x in the form $R \sin(2t - \alpha)$ where $R > 0$ and $0 \leq \alpha \leq \pi/2$. [2]
- ii) Prove that the particle moves in simple harmonic motion. [2]
- iii) Find when the particle is 2m to the right of the origin. [2]
(correct to 2 decimal places)

Question 12 continues on the next page

(d) AB is a tangent to the circle. $AB \parallel EC$ and $CD = AC$.



- Copy the diagram on your answer sheet
- Prove that $AC \parallel ED$

[2]

End of Question 12

Question 13 (Begin a New Page)**(15 Marks)**

(a) The function $f(x)$ is given by $f(x) = \sqrt{x+6}$ for $x \geq -6$

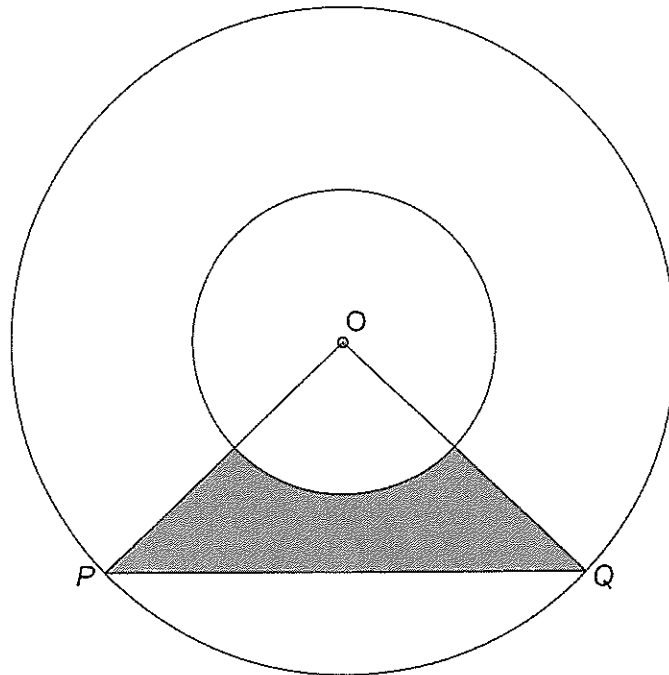
- i) Find the inverse function $f^{-1}(x)$ and find its domain. [2]
- ii) On the same diagram, sketch the graphs of $y = f(x)$ and $y = f^{-1}(x)$. Showing Clearly all the intercepts on the coordinates axes. [2]
- iii) Show that the x coordinates of any points of intersection of the graphs of $y = f(x)$ and $y = f^{-1}(x)$ satisfy the equation $x^2 - x - 6 = 0$. [1]
- iv) Hence find the point of the intersection of the two graphs. [1]

(b) A vertical flagpole CD of height h metres stands with its base C on horizontal ground. A is a point on the ground due west of C and B is a point on the ground 40 metres due south of A . From A and B the angles of elevation of the top D of the flagpole are 20° and 10° respectively.

- i) Draw a diagram for the information given [1]
- ii) Find the height of the flagpole to the nearest metre. [3]

Question 13 continues on the next page

- (c) Two concentric circles with centre O have radii 2 cm and 4 cm . The points P and Q lie on the larger Circle and $\angle POQ = x$, where $0 \leq x \leq \frac{\pi}{2}$



- i) If the area $A\text{ cm}^2$ of the shaded region is $\frac{1}{16}$ the area of the larger circle, show that x satisfies the equation $8\sin x - 2x - \pi = 0$. [1]
- ii) Show that this equation has a solution $x = \alpha$, where $0.5 \leq \alpha \leq 0.6$ [2]
- iii) Taking 0.6 as a first approximation for α , use one application of Newton's method to find a second approximation, giving the answer correct to 2 decimal places. [2]

End of Question 13

Question 14 (Begin a New Page)

(15 Marks)

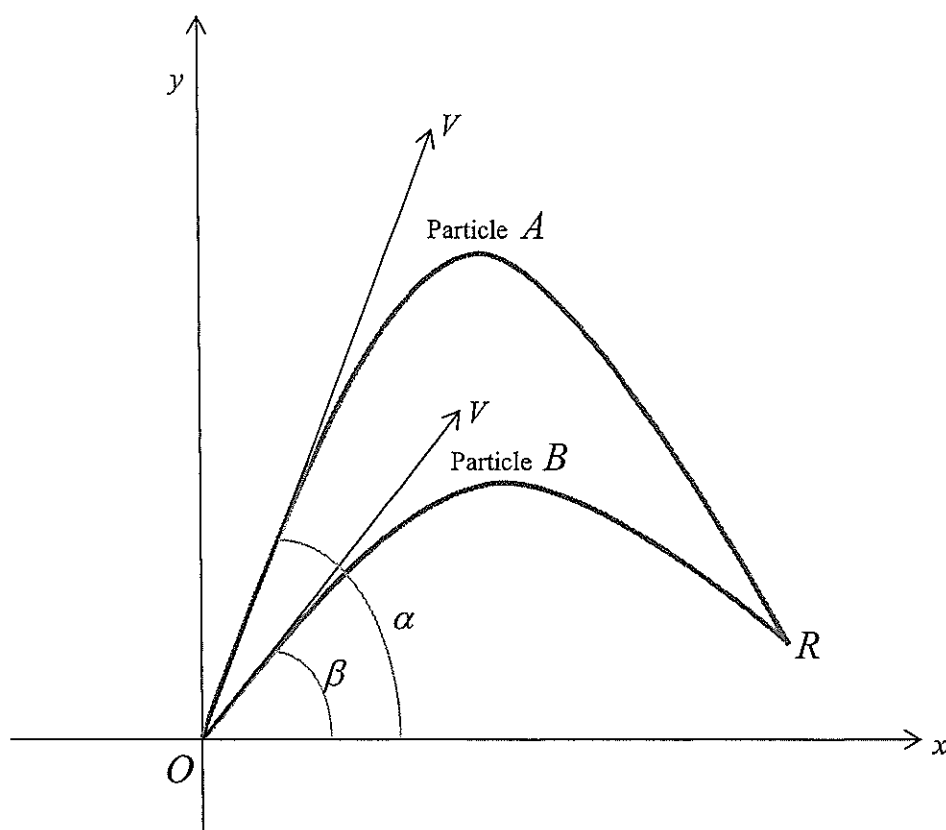
- (a) A particle moves in a straight line. At time t seconds its displacement is x metres from a fixed point O on the line, its acceleration is $a \text{ ms}^{-2}$, and its velocity is $v \text{ ms}^{-1}$, where v is given by $v = \frac{32}{x} - \frac{x}{2}$. Initially the particle is at $x = 2$.

i) Find an expression for a in terms x . [2]

ii) Show that $t = \int \frac{2x}{64 - x^2} dx$, and hence show that $x^2 = 64 - 60e^{-t}$. [3]

iii) Sketch the graph of x^2 against t and describe the limiting behaviour of the particle. [1]

- (b) $P(2t, t^2)$ is a point on the parabola $x^2 = 4y$ with focus F . The point M divides the interval FP externally in the ratio $3:1$. Show that as P moves on the parabola $x^2 = 4y$, then the locus of M is given by $x^2 = 6y + 3$. [3]



- (c) The diagram above shows two particles *A* and *B* projected from the origin. Particle *A* is projected with initial velocity V m/s at an angle α and Particle *B* is projected T seconds later with the same initial velocity V m/s but an angle of β . The particles collide at the point *R*.

- i) You may assume that the equation of the path of *A* is given by

$$y = -\frac{gx^2}{2V^2} \sec^2 \alpha + x \tan \alpha$$

Write down the equation of the path of *B*.

[1]

- ii) Show that the x -coordinate of the collision point *R* is given by

$$x = \frac{2V^2 \cos \alpha \cos \beta}{g \sin(\alpha + \beta)}$$

[2]

- iii) You may assume that the horizontal displacement of *A* after t seconds is given by

$$x = Vt \cos \alpha$$

Write down the equation for the horizontal displacement of *B*.

[1]

- iv) Show that, for the collision to take place, the value of T is given by

$$T = \frac{2V(\cos \beta - \cos \alpha)}{g \sin(\alpha + \beta)}$$

[2]

End of paper

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

Sydney Girls High School Mathematics Faculty

Multiple Choice Answer Sheet – Trial HSC 2013
Extension 1



Student Number: Answers

Completely fill the response oval representing the most correct answer.

1. A ☐ B ☒ C ☐ D ☐
2. A ☐ B ☐ C ☐ D ☒
3. A ☐ B ☐ C ☒ D ☐
4. A ☐ B ☐ C ☐ D ☒
5. A ☐ B ☐ C ☐ D ☒
6. A ☐ B ☒ C ☐ D ☐
7. A ☐ B ☐ C ☒ D ☐
8. A ☒ B ☐ C ☐ D ☐
9. A ☒ B ☐ C ☐ D ☐
10. A ☐ B ☐ C ☐ D ☐ No answer

Section II

Question 11.

a) $u^2 = x+1$
 $u = (x+1)^{1/2}$

$u^2 = x+1$
 $u^2 - 1 = x$

$\frac{du}{dx} = \frac{1}{2}(x+1)^{-1/2}$

$x+2 = u^2 - 1 + 2$
 $= u^2 + 1$

$\frac{du}{dx} = \frac{1}{2\sqrt{x+1}}$

$2du = \frac{dx}{\sqrt{x+1}}$

$\therefore \int \frac{x+2}{\sqrt{x+1}} dx = 2 \int (u^2 + 1) du$

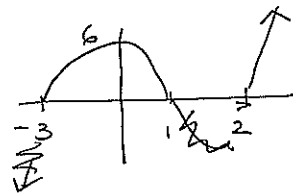
$= 2 \left[\frac{u^3}{3} + u \right] + C$

$= 2 \left[\frac{(x+1)^{3/2}}{3} + (x+1)^{1/2} \right] + C$

$= 2\sqrt{x+1} \left[\frac{(x+1)}{3} + 1 \right] + C$

b) $\frac{(x-1)^2}{x+2} < \frac{4}{(x-1)}$

$(x-1)(x+3)(x-2) < 0$



$\therefore x < -3$

$1 < x < 2$

$(x-1)^2(x+2) < 4(x-1)$

$(x-1)^2(x+2) - 4(x-1) < 0$

$(x-1) \left[(x-1)(x+2) - 4 \right] < 0$

$(x-1) \left[x^2 - x + 2x - 2 - 4 \right] < 0$

$(x-1) \left[x^2 + x - 6 \right] < 0$

c) General solution

$\cos 2x = \cos x$

$\cos 2x - \cos x = 0$

$2\cos^2 x - 1 - \cos x = 0$

$2\cos^2 x - \cos x - 1 = 0$

$(2\cos x + 1)(\cos x - 1) = 0$

$2\cos x = -1$

$\cos x = 1$

$\cos x = -\frac{1}{2}$

$x = \cos^{-1}(1)$

$x = \cos^{-1}\left(-\frac{1}{2}\right)$

$x = 0$

$x = \frac{2\pi}{3} (120^\circ)$

$\therefore x = 2n\pi \pm \frac{2\pi}{3}$

$x = 2n\pi$

d) i) $y = 3 \sin^{-1}\left(\frac{x}{2}\right)$

Domain: $y = \sin^{-1} x \quad -1 \leq x \leq 1$

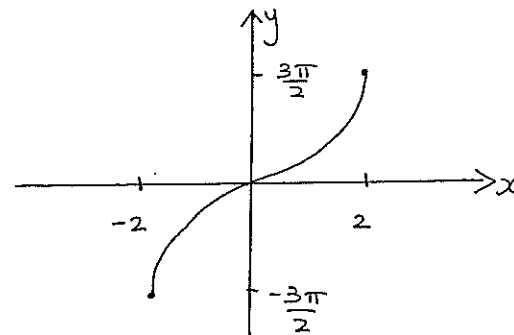
Domain: $y = 3 \sin^{-1} \frac{x}{2} \quad -1 \leq \frac{x}{2} \leq 1$

$-2 \leq x \leq 2$

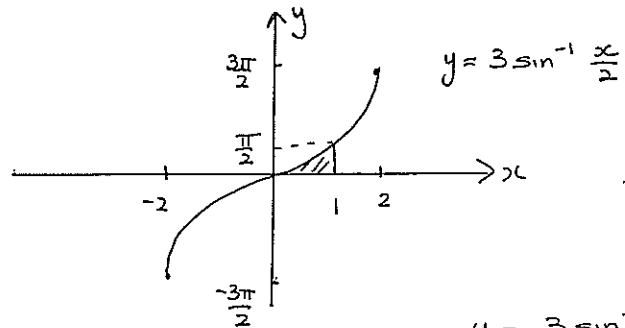
Range: $y = \sin^{-1} x \quad -\frac{\pi}{2} \leq \sin^{-1} x \leq \frac{\pi}{2}$

Range: $y = 3 \sin^{-1} \frac{x}{2} \quad -\frac{3\pi}{2} \leq 3 \sin^{-1} \frac{x}{2} \leq \frac{3\pi}{2}$

$-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$



d) ii)



$$y = 3 \sin^{-1} \frac{x}{2}$$

$$\text{at } x=1, \quad y = 3 \sin^{-1} \frac{x}{2}$$

$$y = 3 \sin^{-1} \frac{1}{2}$$

$$y = \frac{\pi}{2}$$

$$y = 3 \sin^{-1} \frac{x}{2}$$

$$\frac{y}{3} = \sin^{-1} \frac{x}{2}$$

$$\sin\left(\frac{y}{3}\right) = \frac{x}{2}$$

$$2 \sin\left(\frac{y}{3}\right) = x$$

Shaded

$$\text{Area} = \text{Area of rectangle} - \int_0^{\pi/2} 2 \sin \frac{y}{3} dy$$

$$= \left(1 \times \frac{\pi}{2}\right) - 2 \times 3 \left[-\cos \frac{y}{3}\right]_0^{\pi/2}$$

$$= \frac{\pi}{2} - 6 \left[-\cos \frac{\pi}{6} - -\cos 0\right]$$

$$= \frac{\pi}{2} - 6 \left[-\frac{\sqrt{3}}{2} + 1\right]$$

$$= \frac{\pi}{2} + \frac{6\sqrt{3}}{2} - 6$$

$$= \frac{\pi}{2} + 3\sqrt{3} - 6 \text{ units}^2$$

$$\doteq 0.77 \text{ units}^2$$

$$e) \quad i) \quad \tan x + \tan^3 x + \tan^5 x + \dots \quad 0 \leq x \leq \frac{\pi}{4}$$

$$\text{Common ratio} \quad \frac{\tan^3 x}{\tan x} = \frac{\tan^5 x}{\tan^3 x}$$

$$\therefore r = \tan^2 x$$

$$\text{limiting sum exists } -1 < |r| < 1$$

$$\text{if } 0 < x < \frac{\pi}{4} \quad \text{then}$$

$$0 < \tan^2 x < 1 \quad \therefore \text{limiting sum exists}$$

$$ii) \quad S_{\infty} = \frac{a}{1-r}$$

$$= \frac{\tan x}{1 - \tan^2 x}$$

$$= \frac{1}{2} \left(\frac{2 \tan x}{1 - \tan^2 x} \right)$$

$$= \frac{1}{2} \tan 2x$$

Question 12

a) When $n=1$

$5^1 + 2(11^1) = 27$ which is a multiple of 3 ✓

Assume true for $n=k$

$$\frac{5^k + 2(11^k)}{3} = m \text{ (an integer)}$$

$$5^k + 2(11^k) = 3m \Rightarrow 3m - 2(11^k) = 5^k$$

Prove true for $n=k+1$ is

$$5^{k+1} + 2(11^{k+1})$$

$$= 5 \times 5^k + 22 \times 11^k$$

$$= 5(3m - 2(11^k)) + 22 \times 11^k$$

$$= 15m - 10(11^k) + 22 \times 11^k$$

$$= 15m + 12(11^k)$$

$$= 3(5m + 4(11^k))$$

which is a multiple of 3

Hence if true for $n=k$, true for $n=k+1$
True for $n=1$, hence true for $n \geq 1$

b) i) initially $t=0$

$$\text{then } N_A = 15 + 20e^{-0.5t}$$

$$= 35 \quad \checkmark$$

$$\text{ii) } \frac{dN_A}{dt} = -20e^{-0.5t}$$

when $t=0$

$$\frac{dN_A}{dt} = -20 \quad \checkmark$$

$$\text{iii) } N_A = N_B \text{ is } 15 + 20e^{-0.5t} = 5 + 40e^{-0.5t} \quad \checkmark$$

$$10 = 20e^{-0.5t}$$

$$-0.5t = \log_2 \left(\frac{1}{2}\right)$$

$$t = 2 \log_2(2) \quad \checkmark$$

$$= 1.39 \text{ min} \quad (2)$$

$$\text{c) i) } R = \sqrt{6^2 + (2\sqrt{3})^2}$$

$$= \sqrt{48}$$

$$= 4\sqrt{3} \quad \checkmark$$

$$R \sin(2t - \alpha) = 4\sqrt{3} \sin 2t \cos \alpha - 4\sqrt{3} \cos 2t \sin \alpha$$

$$= 6 \sin 2t - 2\sqrt{3} \cos 2t$$

$$\therefore 4\sqrt{3} \cos \alpha = 6 \quad -2\sqrt{3} \sin \alpha = -4\sqrt{3}$$

$$\cos \alpha = \frac{6}{4\sqrt{3}} \quad \sin \alpha = \frac{1}{2} \quad (2)$$

$$\alpha = \frac{\pi}{6} \quad \checkmark$$

$$\text{ii) i) } r = 4\sqrt{3} \sin(2t - \frac{\pi}{6})$$

$$\dot{r} = 8\sqrt{3} \cos(2t - \frac{\pi}{6})$$

$$\dot{r} = -16\sqrt{3} \sin(2t - \frac{\pi}{6})$$

$$= -16r \text{ which is in the form } \dot{r} = -kr$$

iii) when $r=2$

$$4\sqrt{3} \sin(2t - \frac{\pi}{6}) = 2$$

$$\sin(2t - \frac{\pi}{6}) = \frac{1}{2\sqrt{3}}$$

$$2t - \frac{\pi}{6} = \sin^{-1}\left(\frac{1}{2\sqrt{3}}\right)$$

$$t = \frac{\sin^{-1}\left(\frac{1}{2\sqrt{3}}\right) + \frac{\pi}{6}}{2} \quad (2)$$

$$= 0.40822 \dots$$

$$= 0.41 \text{ seconds (must be in radians)}$$

d) Let $\angle DAC = x$

then $\angle ADC = x$ Base $\angle$'s in $\triangle ADC$

then $\angle CAB = \angle ADC$ ($\angle$ in alt segment)

$$= x$$

$$\angle CAB = \angle ACD \text{ Alt } \angle \text{'s } AB \parallel EC$$

$$= x$$

Also $\angle DEB = x$ ($= \angle$'s on chord DC

$\therefore ED \parallel AC$ (equal alt $\angle$'s)

There are many variations on this proof (2)

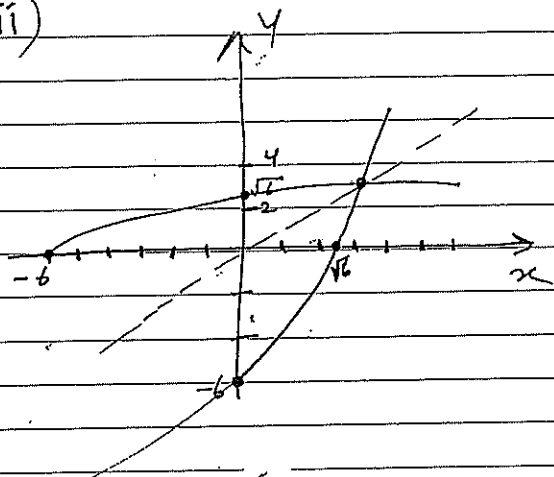
$$13) y = \sqrt{x+6}$$

$$i) x = \sqrt{y+6}$$

$$y = x^2 - 6$$

$$D: x \geq 0$$

ii)



$$iii) x^2 - 6 = x$$

$$x^2 - x - 6 = 0$$

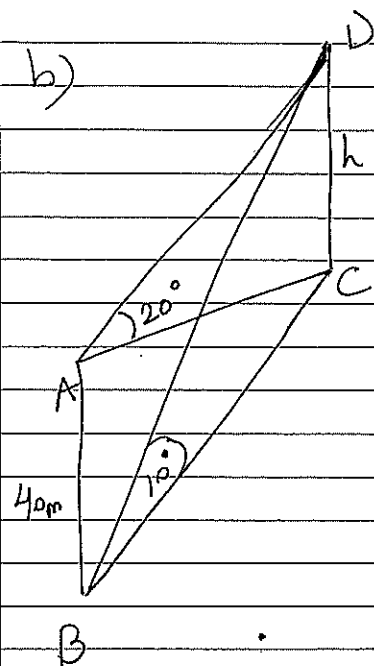
$$iv) x^2 - x - 6 = 0$$

$$(x-3)(x+2) = 0$$

$$x = 3 \quad x \neq -2$$

$$y = 3 \quad P(3, 3)$$

b)



$$\tan 20^\circ = \frac{h}{AC}$$

$$\tan 10^\circ = \frac{h}{BC}$$

$$AC^2 + AB^2 = BC^2$$

$$\left(\frac{h}{\tan 20^\circ}\right)^2 + 40^2 = \left(\frac{h}{\tan 10^\circ}\right)^2$$

$$40^2 = \frac{h^2}{(\tan 10^\circ)^2} - \frac{h^2}{(\tan 20^\circ)^2}$$

$$= \frac{h^2 (\tan 20^\circ)^2 - h^2 (\tan 10^\circ)^2}{(\tan 20^\circ)^2 (\tan 10^\circ)^2}$$

$$h^2 \left[(\tan 20^\circ)^2 - (\tan 10^\circ)^2 \right] = 40^2 (\tan 10^\circ)^2 (\tan 20^\circ)^2$$

$$h = 8 \text{ m}$$

c) i)

$$\frac{1}{2} 4 \times 4 \sin x - \frac{1}{2} \times 2^2 x = \frac{1}{16} \pi \times 16$$

$$8 \sin x - 2x = \pi$$

$$8 \sin x - 2x - \pi = 0$$

ii) $P(0.5) = 8 \sin(0.5) - 1 - \pi$

$$= -0.306$$

$$P(0.6) = 8(\sin(0.6)) - 1 - \pi$$

$$= 0.1755$$

$$P(0.5) < 0, P(0.6) > 0$$

$\therefore x = \alpha$ is a

solution

iii) $x = 0.6 - \frac{P(0.6)}{P'(0.6)}$

$$= 0.6 - \frac{0.1755}{4.603}$$

$$= 0.56$$

14) a)

$$a = \frac{d}{dx} \frac{1}{2} v^2$$

$$\frac{1}{2} v^2 = \frac{1}{2} \left(\frac{32}{x} - \frac{x}{2} \right)^2$$

$$= \frac{1}{2} \left(\frac{1024}{x^2} - 32 + \frac{x^2}{4} \right)$$

$$= \frac{1024}{2x^2} - 16 + \frac{x^2}{8}$$

$$= \frac{512}{x^2} - 16 + \frac{x^2}{8}$$

$$a = -1024x^{-3} + \frac{x}{4}$$

$$= -\frac{1024}{x^3} + \frac{x}{4}$$

ii) $v = \frac{dx}{dt}$

$$v = \frac{32 - x}{2x}$$

$$\frac{dt}{dx} = \frac{2x}{64 - x^2}$$

$$t = \int \frac{2x}{64 - x^2}$$

$$t = - \int \frac{-2x}{64 - x^2}$$

$$t = - \ln(64 - x^2) + C$$

at $t=0$ $x=2$

$$0 = - \ln(60) + C$$

$$C = \ln 60$$

$$t = - \ln(64 - x^2) + \ln 60$$

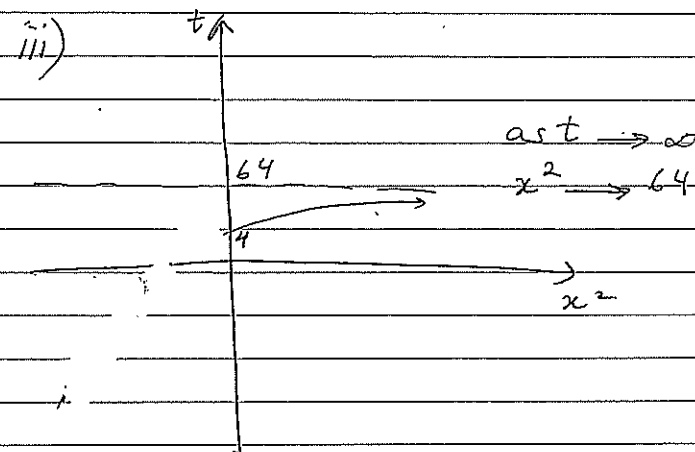
$$-t = \ln(64 - x^2) - \ln 60$$

$$-t = \ln \left(\frac{64 - x^2}{60} \right)$$

$$e^{-t} = \frac{64 - x^2}{60}$$

$$60 e^{-t} = 64 - x^2$$

$$x^2 = 64 - 60 e^{-t}$$



b) $M = \frac{-3 \times 2t + 1 \times 0}{-3 + 1}, \frac{-3t^2 + 1}{-3 + 1}$

$$(x, y) = \frac{-6t}{-2}, \frac{-3t^2 + 1}{-2}$$

$$x = 3t \Rightarrow t = \frac{x}{3}$$

$$y = \frac{-3t^2 + 1}{-2}$$

at $t = \frac{x}{3}$

$$y = \frac{-3 \left(\frac{x}{3} \right)^2 + 1}{-2} = \frac{\frac{-3x^2}{9} + 1}{-2}$$

$$y = \frac{3x^2 - 9}{18} \quad 6y = x^2 - 3 \quad x^2 = 6y + 3$$

i) c) i)

$$y_B = \frac{-gx^2}{2v^2} \sec^2 B + x \tan B$$

$$\text{ii) } \frac{-gx^2}{2v^2} \sec^2 B + x \tan B = \frac{-gx^2}{2v^2} \sec^2 \alpha + x \tan \alpha$$

$$\frac{-gx^2}{2v^2} \sec^2 B + \frac{gx^2}{2v^2} \sec^2 \alpha = x \tan \alpha - x \tan B$$

$$\frac{gx^2}{2v^2} (\sec^2 \alpha - \sec^2 B) = x (\tan \alpha - \tan B)$$

$$\frac{gx}{2v^2} (1 + \tan^2 \alpha - 1 - \tan^2 B) = \tan \alpha - \tan B$$

$$\frac{gx}{2v^2} (\tan \alpha - \tan B) (\tan \alpha + \tan B) = \tan \alpha - \tan B$$

$$\frac{gx}{2v^2} = \frac{1}{\tan \alpha + \tan B}$$

$$x = \frac{2v^2}{g} \cdot \frac{1}{\frac{\sin \alpha}{\cos \alpha} + \frac{\sin B}{\cos B}}$$

$$= \frac{2v^2}{g} \cdot \frac{1}{\frac{\sin \alpha \cos B + \sin B \cos \alpha}{\cos \alpha \cos B}}$$

$$= \frac{2v^2 \cos \alpha \cos B}{g \sin(\alpha + B)}$$

$$\text{iii) } x_B = V(t - T) \cos B$$

$$\text{iv) } \cancel{V} t \cos \alpha = \frac{2V \cancel{V} \cos \alpha \cos B}{g \sin(\alpha + B)}$$

$$V(t - T) \cos B = \frac{2v^2 \cos \alpha \cos B}{g \sin(\alpha + B)}$$

From ①

$$t = \frac{2v \cos B}{g \sin(\alpha + B)}$$

From ②

$$t - T = \frac{2v \cos \alpha}{g \sin(\alpha + B)}$$

$$T = \frac{2v \cos B}{g \sin(\alpha + B)} - \frac{2v \cos \alpha}{g \sin(\alpha + B)}$$

$$= \frac{2v (\cos B - \cos \alpha)}{g \sin(\alpha + B)}$$